Show all working/justify your answers.

(1) Consider the vectors $\mathbf{a} = \langle 5, -3, 4 \rangle$ and $\mathbf{b} = -\mathbf{i} + 3\mathbf{j} + 2\mathbf{k}$.
   (a) Find $\mathbf{a} + \mathbf{b}$.
   (b) Find $\mathbf{a} - \mathbf{b}$.
   (c) Suppose $\mathbf{a}$ and $\mathbf{b}$ are the vectors of two forces being applied to an object. You can balance the forces on the object by applying a third force with vector $\mathbf{c}$. Find $\mathbf{c}$.
   (d) Find $\|\mathbf{a}\|$.
   (e) Find $\mathbf{a} \cdot \mathbf{b}$.
   (f) Are $\mathbf{a}$ and $\mathbf{b}$ parallel, orthogonal, or neither? If neither, is the angle between them acute or obtuse?
   (g) Find $\mathbf{a} \times \mathbf{b}$.
   (h) Find the area of the triangle that has vectors $\mathbf{a}$ and $\mathbf{b}$ as two of its sides.
   (i) Find $\text{proj}_{\mathbf{a}}\mathbf{b}$.
   (j) Find $\text{comp}_{\mathbf{b}}\mathbf{a}$.
   (k) Find the unit vector in the direction of $\mathbf{a}$.

(2) Use vector operations to find the area of the parallelogram with vertices $(1, 1, 1)$, $(-1, -2, 0)$, $(4, 6, 4)$, and $(2, 3, 3)$.

(3) Find the volume of the parallelepiped determined by the vectors $\langle 3, 3, 0 \rangle$, $\langle 5, 4, 3 \rangle$, and $\langle 3, 2, -1 \rangle$.

(4)
   (a) Find a vector that is perpendicular to both $\mathbf{i} - \mathbf{j}$ and $\mathbf{i} + \mathbf{k}$.
   (b) Find a parametric equation of the line that is perpendicular to both $\mathbf{i} - \mathbf{j}$ and $\mathbf{i} + \mathbf{k}$ and passes through the point $(2, -4, 4)$.
   (c) Find a scalar equation for the plane that is parallel to both $\mathbf{i} - \mathbf{j}$ and $\mathbf{i} + \mathbf{k}$ and passes through the point $(0, 3, 1)$.

(5) Consider the quadric surface with equation $\frac{x^2}{4} - \frac{y^2}{9} + z^2 = 1$.
   (a) What kind of surface is this?
   (b) What are the families of traces parallel to each coordinate plane?

(6) Consider the vector-valued function $\mathbf{r}(t) = \left\langle t^2 + 1, 2t + 1, t^4 \right\rangle$.
   (a) Find the derivative $\mathbf{r}'(t)$.
   (b) Consider the space curve you obtain when you plot $\mathbf{r}(t)$; find a parametric equation for the tangent line to that curve at the point corresponding to $t = 2$.
   (c) Find the indefinite integral $\int \mathbf{r}(t)\,\mathrm{d}t$.
   (d) Find the definite integral $\int_0^1 \mathbf{r}(t)\,\mathrm{d}t$.

(7) Let $\mathbf{u}$ and $\mathbf{v}$ be a pair of three-dimensional vectors.
   (a) Prove that $\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u})$.
   (b) Prove that $\mathbf{u} \times \mathbf{v}$ is orthogonal to $\mathbf{u}$.

Formula sheet: trig identities.

$$\sin^2(x) + \cos^2(x) = 1$$
$$\sin(0) = \sin(2\pi) = 0$$
$$\sin\left(\frac{\pi}{2}\right) = 1$$
$$\sin(\pi) = 0$$
$$\sin\left(\frac{3\pi}{2}\right) = -1$$
$$\cos(0) = \cos(2\pi) = 1$$
$$\cos\left(\frac{\pi}{2}\right) = 0$$
$$\cos(\pi) = -1$$
$$\cos\left(\frac{3\pi}{2}\right) = 0$$